

```

def lps(s):
    n = len(s)

    dp = [[0] * n for _ in range(n)]

    for i in range(n):
        dp[i][i] = 1

    for length in range(2, n + 1):
        for i in range(n - length + 1):

            j = i + length - 1

            if s[i] == s[j]:
                if length == 2:
                    dp[i][j] = 2
                else:
                    dp[i][j] = dp[i + 1][j - 1] + 2
            else:
                dp[i][j] = max(dp[i + 1][j],
                               dp[i][j - 1])

    return dp[0][n - 1]

s = input("Enter string: ")

print("Length of Longest Palindromic Subsequence:",
      lps(s))

```

Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>

==== RESTART: C:/Users/ROOPA SAI DIVYA/OneDrive/Documents/Big-O, Ω and Θ .py ====

Enter string: BBABCBAB

Length of Longest Palindromic Subsequence: 7

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